Q.5 What is an electric dipole? Calculate electric field at a point due to electric dipole?

ELECTRIC DIPOLE

A pair of positive charge and negative charge of same magnitude having constant distance between them is called electric dipole. It is a useful concept in atoms and molecules where the effects of charge separation are measurable. It is also a useful concept in dielectrics and other applications in solid and liquid materials.

ELECTRIC DIPOLE MOMENT

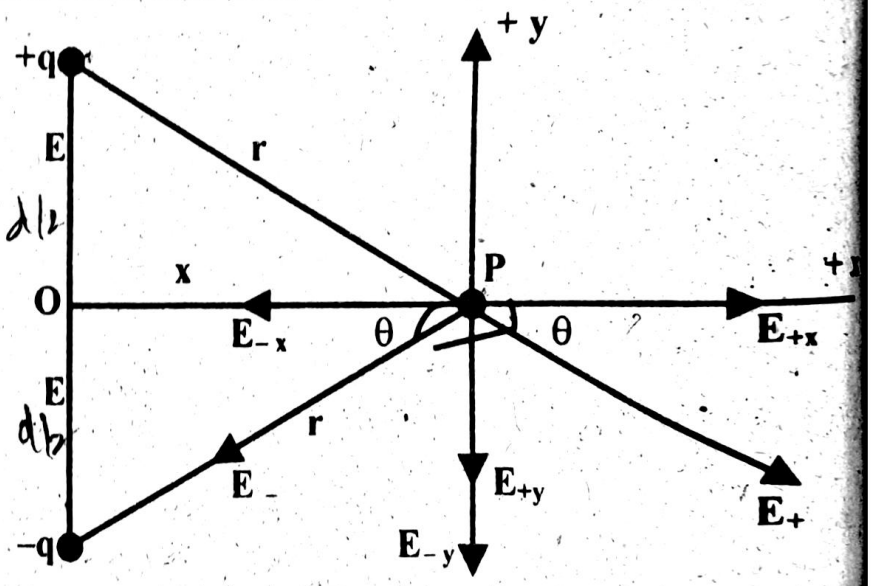
The product of magnitude of charge and separation between them is called electric dipole moment and written as $P = q\,d$.

The electric dipole moment is a vector quantity. The direction of electric dipole moment vector is from negative charge to positive charge.

ELECTRIC FIELD DUE TO ELECTRIC DIPOLE

Consider a +q and −q charge having distance d between them. This is called electric dipole. The dipole moment is given as $P = qd$

Take a point P having distance 'x' which is perpendicular bisector of distance d as shown in fig.



The magnitude of electric field at P due to +q charge having distance r is

$$E_+ = \frac{kq}{r^2} \qquad \text{-----(1)}$$

The direction of E_+ is from +q charge to towards P. The magnitude of electric field at P due to −q charge having distance r is

$$E_- = \frac{kq}{r^2} \qquad \text{---------(2)}$$

The direction of E_- is from P to towards −q charge.

Comparing eq(1) and eq(2) we can write as

$$\boxed{E_+ = E_-}$$

Now to calculate net electric field at P due to dipole, resolve the electric field E_+ and E_- into components.

Rectangular components of E_+ are

$$E_{+x} = E_+ \cos\theta \quad , \quad E_{+y} = E_+ \sin\theta$$

Rectangular components of E_- are

$$E_{-x} = E_- \cos\theta \quad , \quad E_{-y} = E_- \sin\theta$$

The resultant x –component of electric field is

$$E_x = (+E_{+x}) + (-E_{-x})$$

$$E_x = E_+ \cos\theta - E_- \cos\theta$$

$$E_x = E_+ \cos\theta - E_+ \cos\theta$$

$$E_x = 0$$

The resultant y –component of electric field is

$E_{+y} \sin\theta - E_{-y} \sin$

$$E_y = (-E_{+y}) + (-E_{-y})$$

$E_{+y} \sin\theta + E$

$$E_y = -E_+ \sin\theta - E_- \sin\theta$$

due lin on 3rd Axis

$$E_y = -E_+ \sin\theta - E_+ \sin\theta$$

$$E_y = -2E_+ \sin\theta$$ → y-component lie on -axis

Magnitude of resultant electric field E is

$$E = |E_x| + |E_y| = 0 + (-2E_+ \sin\theta)$$

$$E = 2E_+ \sin\theta$$

$$E = 2\frac{kq}{r^2}\sin\theta$$

$$E = 2\frac{kq}{r^2}\frac{(d/2)}{r} \qquad \left(\text{From fig. } \sin\theta = \frac{d/2}{r}\right)$$

$$E = \frac{k(qd)}{r^3} = \frac{kP}{r^3}$$

$$E = kPr^{-3}$$

$$E = kP\left(x^2 + \frac{d^2}{4}\right)^{-3/2} \qquad \left(\text{From fig. } r^2 = x^2 + \frac{d^2}{4}\right)$$

$$E = kPx^{-3}\left(1 + \frac{d^2}{4x^2}\right)^{-3/2} = \frac{kP}{x^3}\left(1 + \frac{d^2}{4x^2}\right)^{-3/2}$$

Expand the term in bracket by binomial theorem

$$E = \frac{kP}{x^3}\left[1 - \left(\frac{3}{2}\right)\frac{d^2}{4x^2} + \text{--------}\right]$$

Neglect the higher order terms, we get

$$E = \frac{kP}{x^3}$$

This is the electric field at P due to electric dipole. Where x is distance between point p and bisecting point O of distance between +q and –q.